Assignment 7

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Q1

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[4] I. Let $f(x) = \arctan(x^2)$. On the interval $[-1, 1]$, find a constant B such that

$$|f(x) - T_{2,0}(x)| = |R_{2,0}(x)| \leq B\,|x|^3.$$

$$f'(x) = \frac{2x}{1+x^4}$$

$$f''(x) = \frac{-4x^3(2x)}{(1+x^4)^2} + \frac{2}{1+x^4} = \frac{2+2x^4}{(1+x^4)^2} - \frac{8x^4}{(1+x^4)^2} = \frac{2-6x^4}{(1+x^4)^2}$$

$$f'''(x) = \frac{-2(4x^3)(2-6x^4)}{(1+x^4)^3} + \frac{-24x^3}{(1+x^4)^2} = \frac{(8x^3)(6x^4-2)}{(1+x^4)^3} + \frac{-24x^3-24x^7}{(1+x^4)^3} = \frac{24x^7-40x^3}{(1+x^4)^3}$$

$$|R_{n,a}(x)| \leq K\frac{|x-a|^{n+1}}{(n+1)!}, \quad \text{where } |f^{(n+1)}(x)| \leq K$$

$$|f'''(x)| = \left|\frac{8x^3(3x^4-5)}{(1+x^4)^3}\right|$$

For $x \in [-1, 1]$; $|8x^3| \leq 8$, $|3x^4-5| \leq 5$, $|(1+x^4)^3| \geq 1$

So $|f'''(x)| \leq 40 \implies |R_{2,0}(x)| \leq B|x^3|$ where $\boxed{B = \frac{40}{3!} = \frac{20}{3}}$

Q2

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[4] II. Let $f(x) = e^x$. How large must n be so that $T_{n,0}(x)$ is not to exceed an error of 10^{-4} on $x \in [-1, 1]$?

$$T_{n,0}(x) = \sum_{k=0}^{n} \frac{x^n}{n!}$$

$$|R_n| \leq K \frac{|x|^{n+1}}{(n+1)!},$$

$$K = \max_{x\in[-1,1]} \left\{ \left| \frac{d^{n+1}}{dx^{n+1}} e^x \right| \right\} = \max_{x\in[-1,1]} \{|e^x|\} = e$$

$$|R_n| \leq e \frac{|x|^{n+1}}{(n+1)!}, \quad x \in [-1,1]$$

$$|R_n| \leq \frac{e}{(n+1)!}. \quad \text{Let } \frac{e}{(n+1)!} < 10^{-4} \Rightarrow (n+1)! > \frac{e}{10^{-4}}$$

$$(6+1)! = 5040 < e \cdot 10^4$$
$$(7+1)! = 40320 > e \cdot 10^4$$

Therefore $\boxed{n \geq 7}$

Q3

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[4] III. Evaluate $\lim_{x \to 0} \frac{2\sin x - \sin(2x)}{2e^x - 2 - 2x - x^2}$ using Taylor polynomials of degree 5.

Expression $= \lim_{x \to 0} \frac{2\cos x - 2\cos(2x)}{2e^x - 2 - 2x} = \lim_{x \to 0} \frac{-2\sin x + 4\sin(2x)}{2e^x - 2} = \lim_{x \to 0} \frac{8\cos(2x) - 2\cos x}{2e^x} = 3$

Let $f(x) = 2\sin x - \sin(2x)$, $g(x) = 2e^x - 2 - 2x - x^2$, $F(x) = \frac{f(x)}{g(x)}$

Re-write equation: $\lim_{x \to 0} \frac{f(x)}{g(x)} = \lim_{x \to 0} \frac{f'(x)}{g'(x)} = \lim_{x \to 0} \frac{f''(x)}{g''(x)} = \lim_{x \to 0} \frac{f'''(x)}{g'''(x)} = \frac{6}{2}$

$Tf_{n,0}(x) = \sum_{k=0}^{n} \frac{f^{(k)}(0)}{k!} x^k$, where $f(0) = f'(0) = f''(0) = 0$, $f'''(0) = 6$

$Tg_{n,0}(x) = \sum_{k=0}^{n} \frac{g^{(k)}(0)}{k!} x^k$, where $g(0) = g'(0) = g''(0) = 0$, $g'''(0)$

By construction, $\forall i \in [0, n]$, $Tf_{n,0}^{(i)}(0) = f^{(i)}(0)$ and T

Therefore, $\forall n \geq 3$: $\lim_{x \to 0} \frac{Tf_{n,0}(x)}{Tg_{n,0}(x)} = \lim_{x \to 0} \frac{[Tf_{n,0}(x)]'}{[Tg_{n,0}(x)]'}$

you did not show these computations **−1**

you did not use Taylor polynomials of degree 5 to evaluate the limit. Please see solutions **−2**

Q4

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[6] IV. Let $I = \int_0^{1/2} e^{-x^2}\,dx$.

(a) Find $T_{n,0}(x)$ for $f(x) = e^x$ and deduce the n-th degree Taylor polynomial centerd at 0 for e^{-x^2}.

(b) Given that $f(x) = T_{n,0}(x) + R_{n,0}(x)$, use Taylor's inequality to prove that

$$R_{n,0}(-x^2) \le \frac{x^{2n+2}}{(n+1)!},$$

for $x \in \left[0, \frac{1}{2}\right]$.

(c) Use parts (a) and (b) to estimate I within an error less that $\frac{1}{1000}$.

(a) Since $(e^x)^{(n)} = e^x$, $T_{n,0}(x) = \sum_{k=0}^{n} \frac{x^k}{k!}$ (for e^x)

Take $u = -x^2$ into expression: $e^{-x^2} = T_{n,0}(-x^2) = \boxed{\sum_{k=0}^{n} (-1)^k \frac{x^{2k}}{k!}}$

part (a) 1

(b) Taylor's inequality: $|R_{n,a}(x)| \le K \frac{|x-a|^{n+1}}{(n+1)!}$, where $K \le f^{(n+1)}(x)$

$|R_{n,0}(-x^2)| \le K \frac{|(-x)^{2n+2}|}{(n+1)!} = \boxed{K \frac{x^{2n+2}}{(n+1)!}}$

where $K = \max_{x \in [0,\frac{1}{2}]} \left\{ f^{(n+1)}(-x^2) \right\} = \max_{x \in [0,\frac{1}{2}]} \left\{ e^{-x^2} \right\} =$

part (b) 2

(c) Let $F(x) = \int_0^x e^{-t^2}\,dt$, $I = F(\frac{1}{2})$

$F(x) \approx \int_0^x T_{n,0}(-t^2)\,dt = \int_0^x \sum_{k=0}^{n} (-1)^k \frac{t^{2k}}{k!}\,dt = \sum_{k=0}^{n} (-1)^k \frac{x^{2k+1}}{k!(2k+1)}$

Using the Alternating Series Estimation Theorem: $|S - S_n| \le a_{n+1}$

Error $\le \frac{x^{2n+3}}{(n+1)!(2n+3)}$, $(x = \frac{1}{2})$ let Error $< \frac{1}{1000}$, $\boxed{n \ge 2}$

So $I = F(\frac{1}{2}) \approx \sum_{k=0}^{2} (-1)^k \frac{x^{2k+1}}{k!(2k+1)} \approx \boxed{0.4615}$ with Error $< \frac{1}{1000}$

part (c) 3